5-state EKF controller (disturbance state removed). The combined disturbance state δx_D^d of the 6-state baseline is dropped, so the augmented per-axis state is $\mathbf{x} = [\,\delta x_1,\ \delta x_2,\ \delta x_3,\ a_x,\ \delta a_x\,]^\mathsf{T}$. Plant, Measurement, Control objective and the *ideal* control effort are identical to the 6-state baseline (`6state_derivation_xd.tex`) and are not reprinted; the divergences appear only in the *implementable* law and everything downstream of it.

Implementable control effort

(5-state: the $-\,\delta\hat{x}_D^d[k]$ term of the 6-state law is absent — the disturbance state is removed, so there is nothing to subtract.)

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k]\big[f_{dx}[k] - \bar{\bar{f}}_{dx}[k]\big] - \delta x_T^d[k]$$

$$a_x[k]\,\bar{\bar{f}}_{dx}[k] = \Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big] - \delta x_D^d[k]\Big\}$$

$$\begin{aligned} a_x[k]\,f_{dx}[k] = \big(1 + e_{a_x}[k]\,\hat{a}_x^{-1}[k]\big)\Big\{&\Delta x_d^d[k] \\ &+ (1-\lambda_c)\Big[\big(\delta x[k-d] + n_x[k]\big) - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\} \end{aligned}$$

(5-state: no $\delta\hat{x}_D^d$ in the brace $\Rightarrow$ the difference leaves the uncompensated $\delta x_D^d[k]$ below, where the 6-state had $e_{\delta x_D^d}[k]$.)

$$\begin{aligned} a_x[k]\,f_{dx}[k] - a_x[k]\,\bar{\bar{f}}_{dx}[k] = {}& e_{a_x}[k]\,f_{dx}[k] + (1-\lambda_c) n_x[k] \\ &+ (1-\lambda_c)\textstyle\sum_{i=1}^{d}\big(a_x[k-i] - \hat{a}_x[k-i]\big) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model:

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a_x'[k]\,x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= \boldsymbol{a_{x_{det}}[k+1]} + a_{x_{ram}}[k+1] \\ &= \underbrace{\Big\{\boldsymbol{a_{x_{det}}[k]} + a_{x_{ram}}[k]\Big\}}_{a_x[k]} + \boldsymbol{\delta a[k]} + \underbrace{\Big\{a_{x_{ram}}[k+1] - a_{x_{ram}}[k]\Big\}}_{\delta a_{ram}[k]} \end{aligned}$$

$$\begin{cases} a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k] \\ \delta a_x[k+1] = \delta a_x[k] \end{cases}$$

Parameter estimator:

$$\begin{cases} \hat{a}_x[k+1] = \hat{a}_x[k] + \delta\hat{a}_x[k] \\ \delta\hat{a}_x[k+1] = \delta\hat{a}_x[k] \end{cases}$$

Parameter estimation error:

$$\begin{cases} e_{a_x}[k] = a_x[k] - \hat{a}_x[k] \\ e_{\delta a_x}[k] = \delta a_x[k] - \delta\hat{a}_x[k] \end{cases}$$

Estimation error dynamics:

$$\begin{cases} e_{a_x}[k+1] = e_{a_x}[k] + e_{\delta a_x}[k] + \delta a_{ram}[k] \\ e_{\delta a_x}[k+1] = e_{\delta a_x}[k] \end{cases}$$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - i \cdot e_{\delta a_x}[k] - \sum_{j=1}^{i} \delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{\bar{f}}}_{dx}[k] \approx\; & e_{a_x}[k]\, \underbrace{\left\{ f_{dx}[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ & - e_{\delta a_x}[k]\, \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} i \cdot f_{dx}[k-i] \right\}}_{dF_{dx}[k]} \\ & - \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} (d+1-i) \cdot f_{dx}[k-i]\, \delta a_{ram}[k-i] \right\}}_{\delta x^{d}_{\delta af}[k]} \\ & + \delta x^{d}_{D}[k] + (1-\lambda_c) n_x[k] \end{aligned}$$

(5-state: $\delta x^{d}_{D}[k]$ is the uncompensated disturbance, not an estimation error; the EKF models it as 0 ($\delta x_D \approx 0$) and the state equation below drops it.)

$$\boxed{\begin{aligned} \delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - \delta x^{d}_{D}[k] \\ - \underbrace{\left\{ \delta x^{d}_{T}[k] - \delta x^{d}_{\delta af}[k] + (1-\lambda_c) n_x[k] \right\}}_{\varepsilon[k]} \end{aligned}}$$

State equation

(EKF model: $\delta x^{d}_{D} \equiv 0$, state removed)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k]$$

$$\delta a_x[k+1] = \delta a_x[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} \approx a_x[k] - d \cdot \delta a_x[k] + \underbrace{\left\{ n_a[k] - \textstyle\sum_{i=1}^{d} \delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -d \end{bmatrix}$$

$$\mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

(5-state: the $\delta\hat{x}_D^d$ update line is removed; the gain rows for $\hat{a}_x$, $\delta\hat{a}_x$ shift up to $\ell_{4\bullet}$, $\ell_{5\bullet}$.)

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta\hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{\delta a_x}[k] = \delta a_x[k] - \delta\hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \hat{a}_x[k] + d\cdot\delta\hat{a}_x[k] \\ &= e_{a_x}[k] - d\cdot e_{\delta a_x}[k] + \underbrace{\left\{n_a[k] - \textstyle\sum_{i=1}^{d}\delta a_{ram}[k-i]\right\}}_{r_2[k]} \end{aligned}$$

$$\begin{aligned} \delta\hat{x}_1[k+1] &= \delta\hat{x}_2[k] + \ell_{11}[k]\,e_{y_1}[k] + \ell_{12}[k]\,e_{y_2}[k] \\ \delta\hat{x}_2[k+1] &= \delta\hat{x}_3[k] + \ell_{21}[k]\,e_{y_1}[k] + \ell_{22}[k]\,e_{y_2}[k] \\ \delta\hat{x}_3[k+1] &= \lambda_c\cdot\delta\hat{x}_3[k] + \ell_{31}[k]\,e_{y_1}[k] + \ell_{32}[k]\,e_{y_2}[k] \\ \hat{a}_x[k+1] &= \hat{a}_x[k] + \delta\hat{a}_x[k] + \ell_{41}[k]\,e_{y_1}[k] + \ell_{42}[k]\,e_{y_2}[k] \\ \delta\hat{a}_x[k+1] &= \delta\hat{a}_x[k] + \ell_{51}[k]\,e_{y_1}[k] + \ell_{52}[k]\,e_{y_2}[k] \end{aligned}$$

Error dynamics

$$\begin{aligned} e_{\delta x_1}[k+1] &= e_{\delta x_2}[k] - \ell_{11}[k]\,e_{\delta x_1}[k] - \ell_{12}[k]\,e_{a_x}[k] + \ell_{12}[k]\cdot d\cdot e_{\delta a_x}[k] \\ e_{\delta x_2}[k+1] &= e_{\delta x_3}[k] - \ell_{21}[k]\,e_{\delta x_1}[k] - \ell_{22}[k]\,e_{a_x}[k] + \ell_{22}[k]\cdot d\cdot e_{\delta a_x}[k] \\ e_{\delta x_3}[k+1] &= \lambda_c\cdot e_{\delta x_3}[k] - F_{dx}[k]\cdot e_{a_x}[k] + dF_{dx}[k]\cdot e_{\delta a_x}[k] \\ &\qquad - \ell_{31}[k]\,e_{\delta x_1}[k] - \ell_{32}[k]\,e_{a_x}[k] + \ell_{32}[k]\cdot d\cdot e_{\delta a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]} \\ e_{a_x}[k+1] &= e_{a_x}[k] + e_{\delta a_x}[k] - \ell_{41}[k]\,e_{\delta x_1}[k] - \ell_{42}[k]\,e_{a_x}[k] + \ell_{42}[k]\cdot d\cdot e_{\delta a_x}[k] + \underbrace{\delta a_{ram}[k]}_{q_4[k]} \\ e_{\delta a_x}[k+1] &= e_{\delta a_x}[k] - \ell_{51}[k]\,e_{\delta x_1}[k] - \ell_{52}[k]\,e_{a_x}[k] + \ell_{52}[k]\cdot d\cdot e_{\delta a_x}[k] \end{aligned}$$

(5-state: the 6-state's $e_{\delta x_D^d}$ line is gone, and the $-e_{\delta x_D^d}[k]$ term that sat in the $e_{\delta x_3}$ row is dropped with the disturbance state. Process noise: $q_3 = \varepsilon$, $q_4 = \delta a_{ram}$.)

(5-state: 5×5. Row 3 loses the 6-state col-4 entry -1 that coupled $\delta x_D^d \to \delta x_3$; the gain states a_x, δa_x occupy cols 4,5.)

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF_{dx}[k] \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^{d} f_{dx}[k-i], \qquad dF_{dx}[k] = (1 - \lambda_c) \sum_{i=1}^{d} i \cdot f_{dx}[k-i]$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}\,\mathbf{r}[k]$$